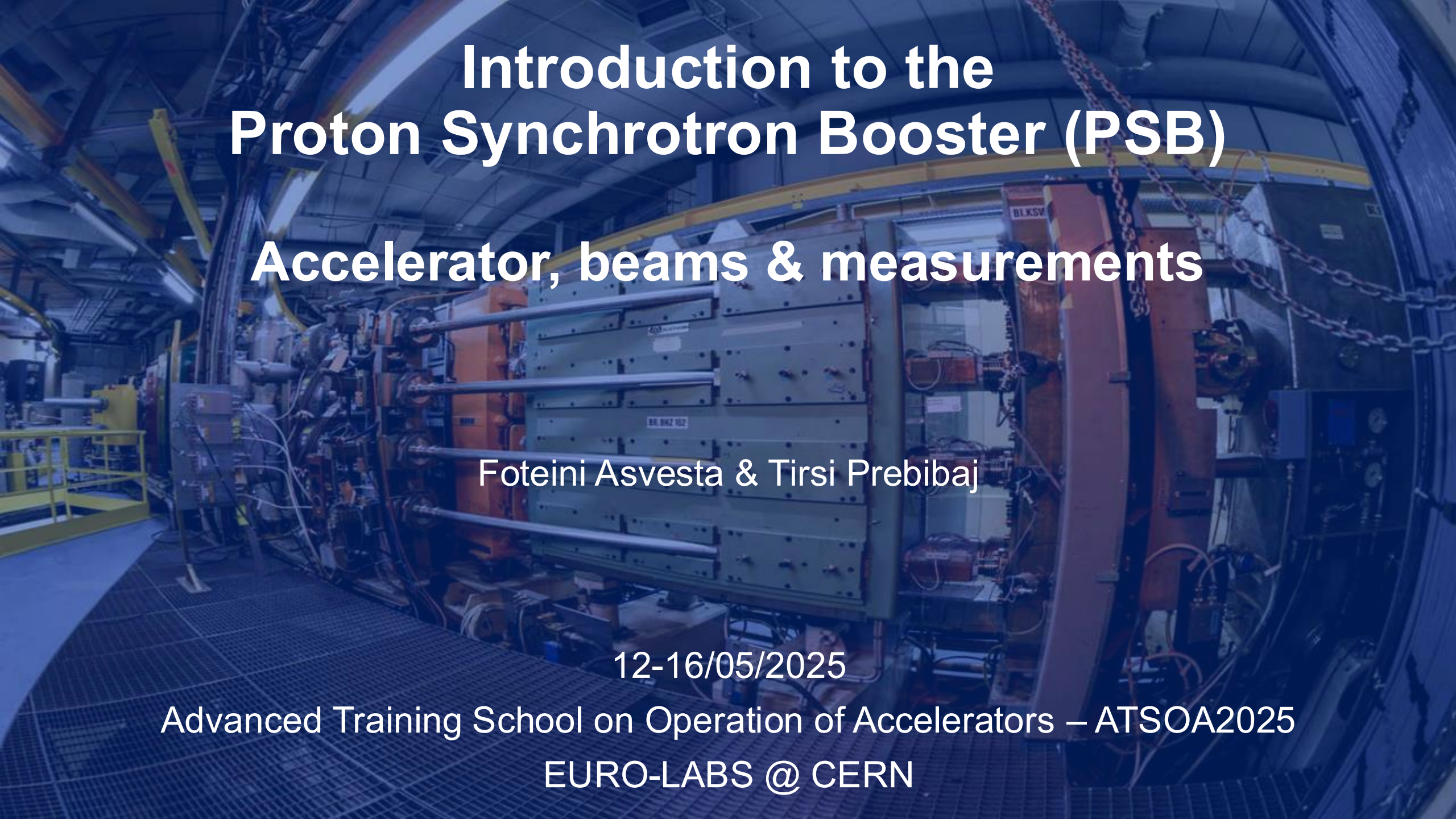






Introduction to the Proton Synchrotron Booster (PSB)

Accelerator, beams & measurements

Foteini Asvesta & Tirsi Prebibaj

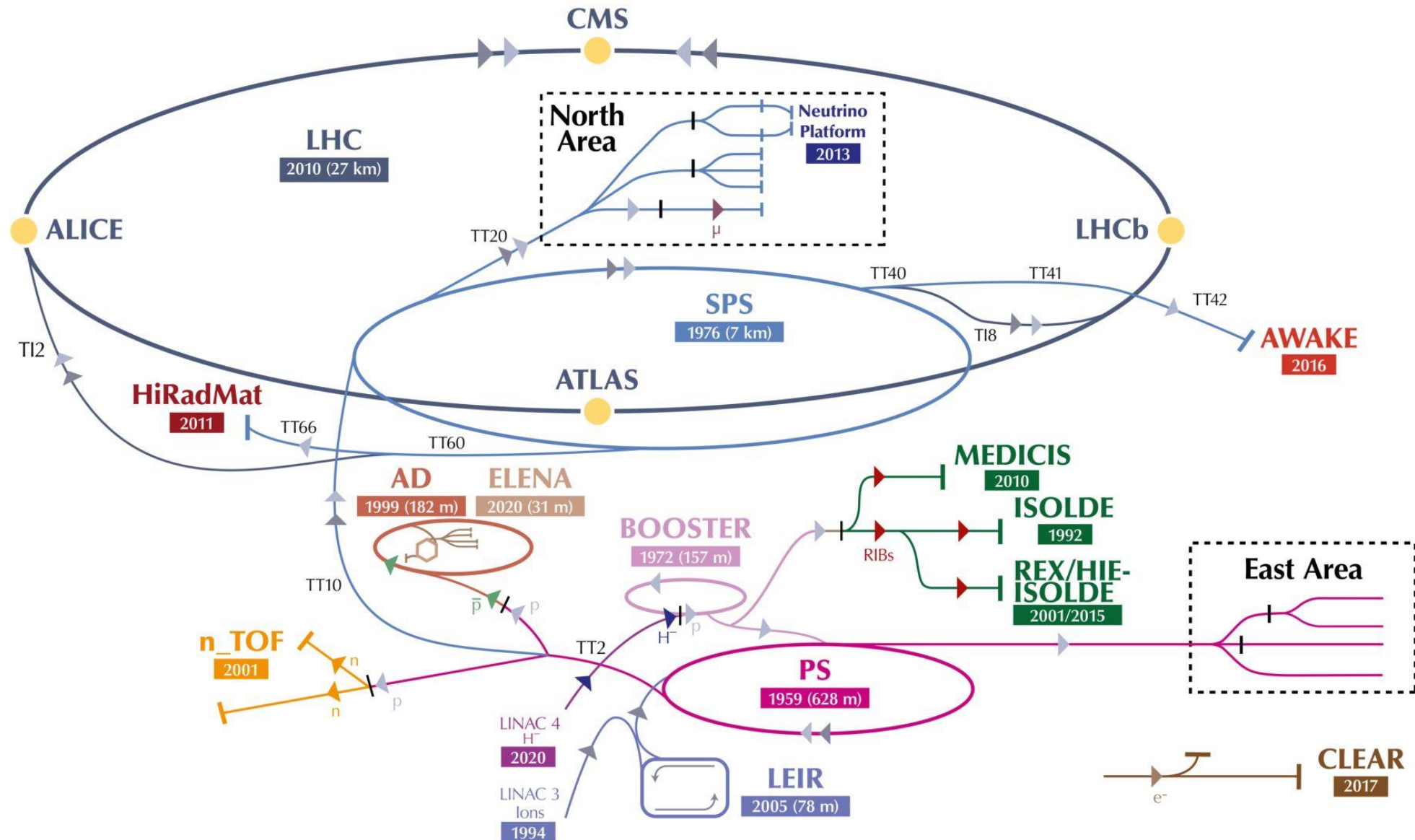
12-16/05/2025

Advanced Training School on Operation of Accelerators – ATSOA2025
EURO-LABS @ CERN

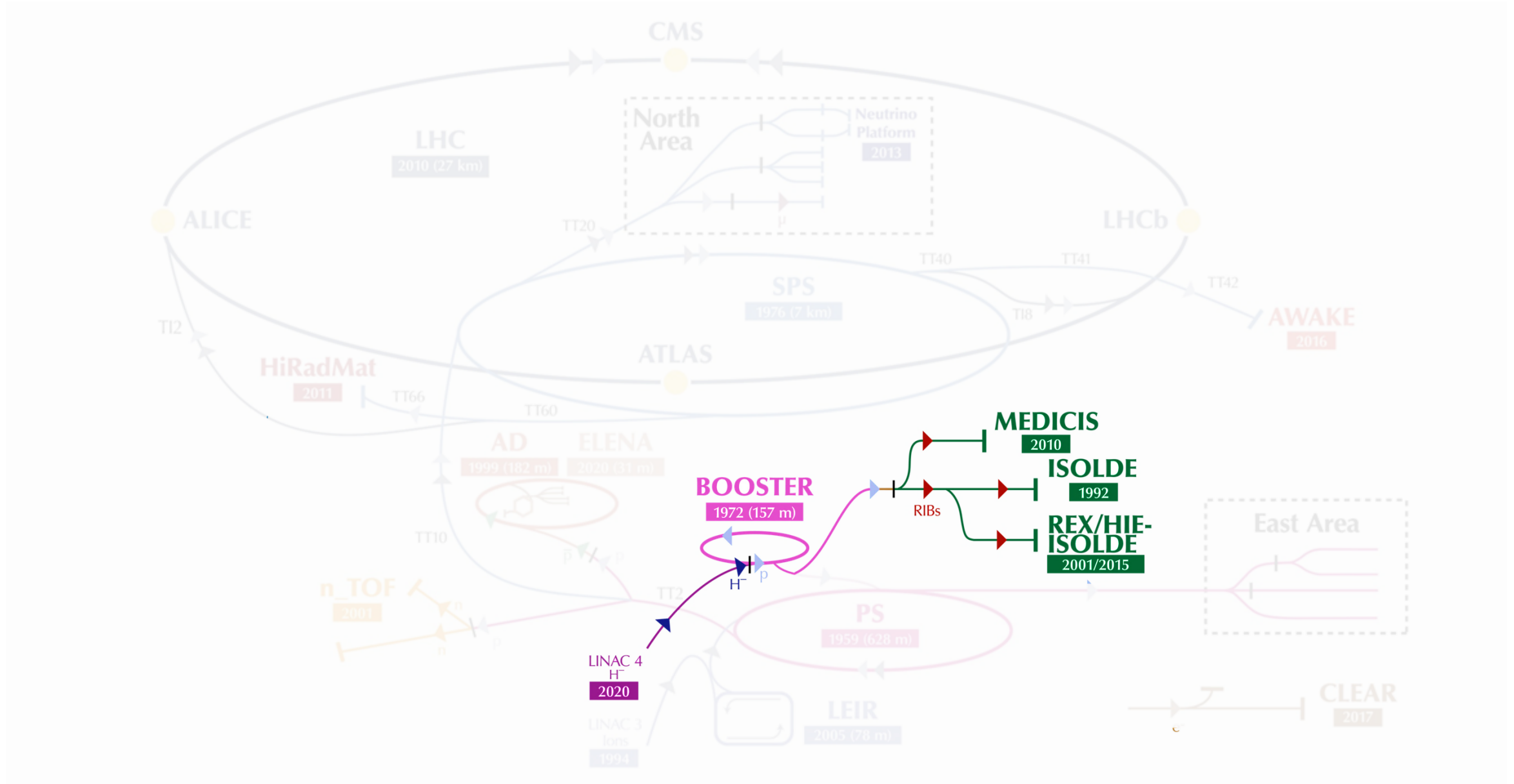
Outline

- CERN injectors complex & facilities
- PSB layout
- Beam delivery – supercycles
- Beam instrumentation
- Beam dynamics
- Data acquisition
- Measurements for today
- Space charge & resonances

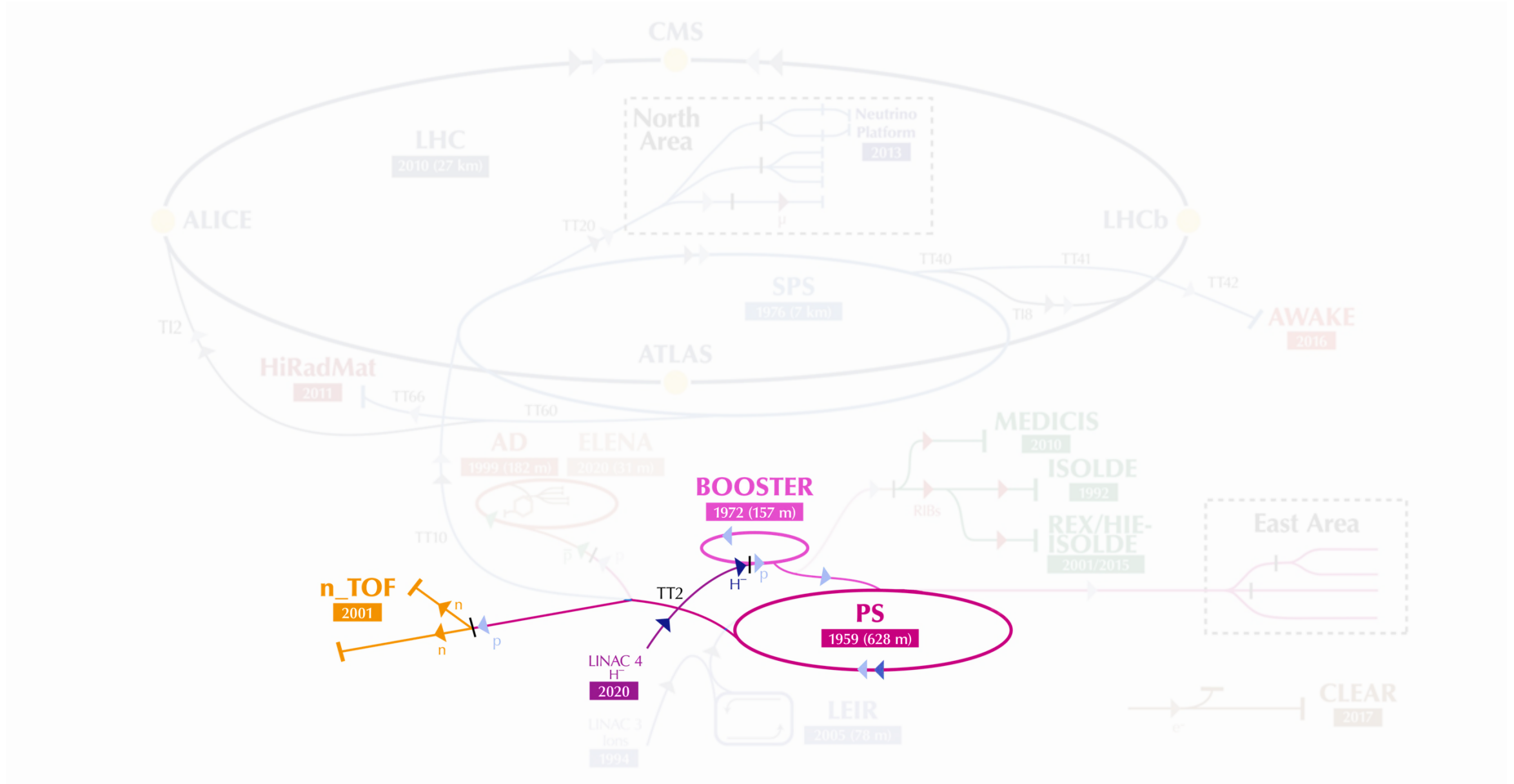
CERN accelerator complex & facilities



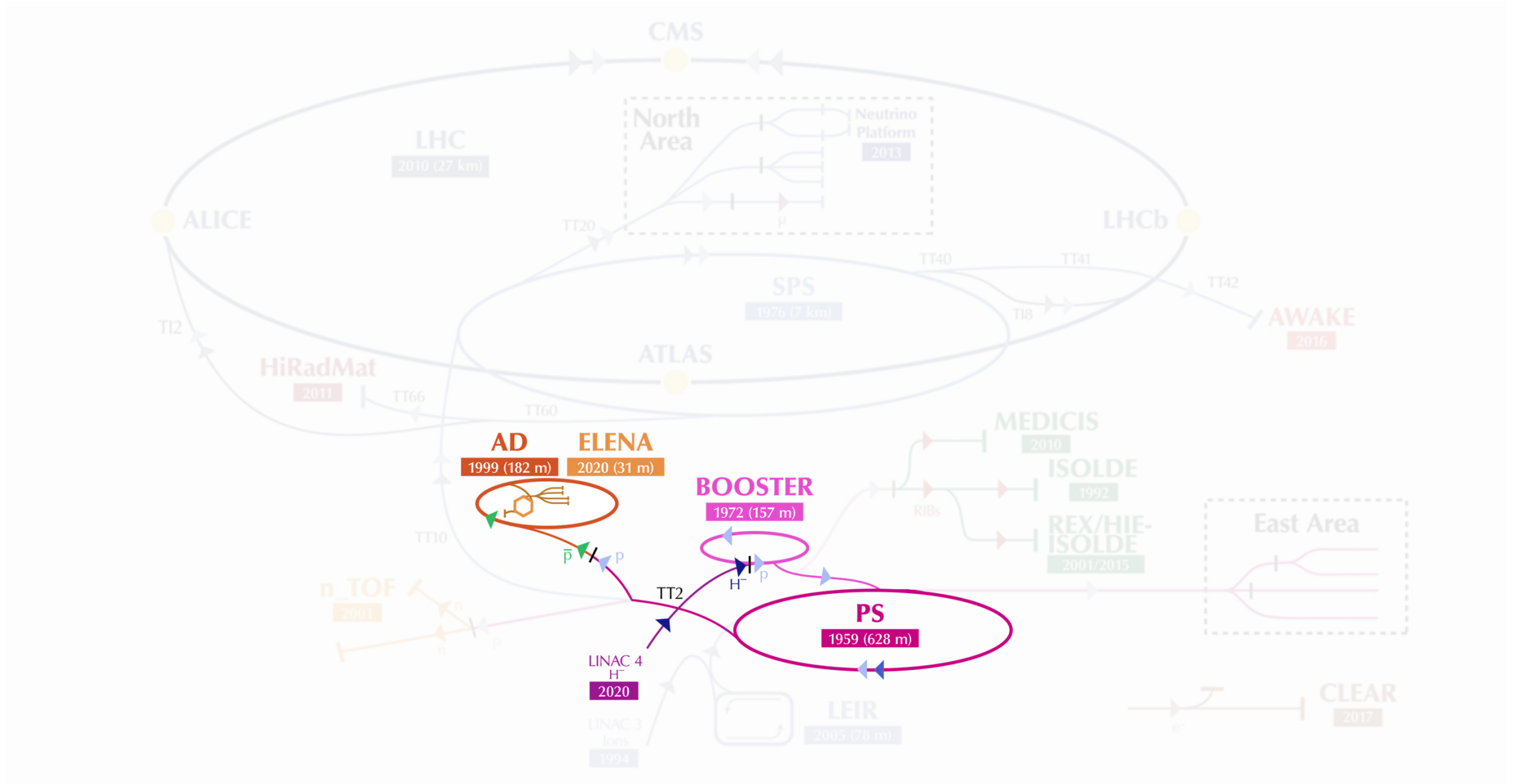
CERN accelerator complex & facilities



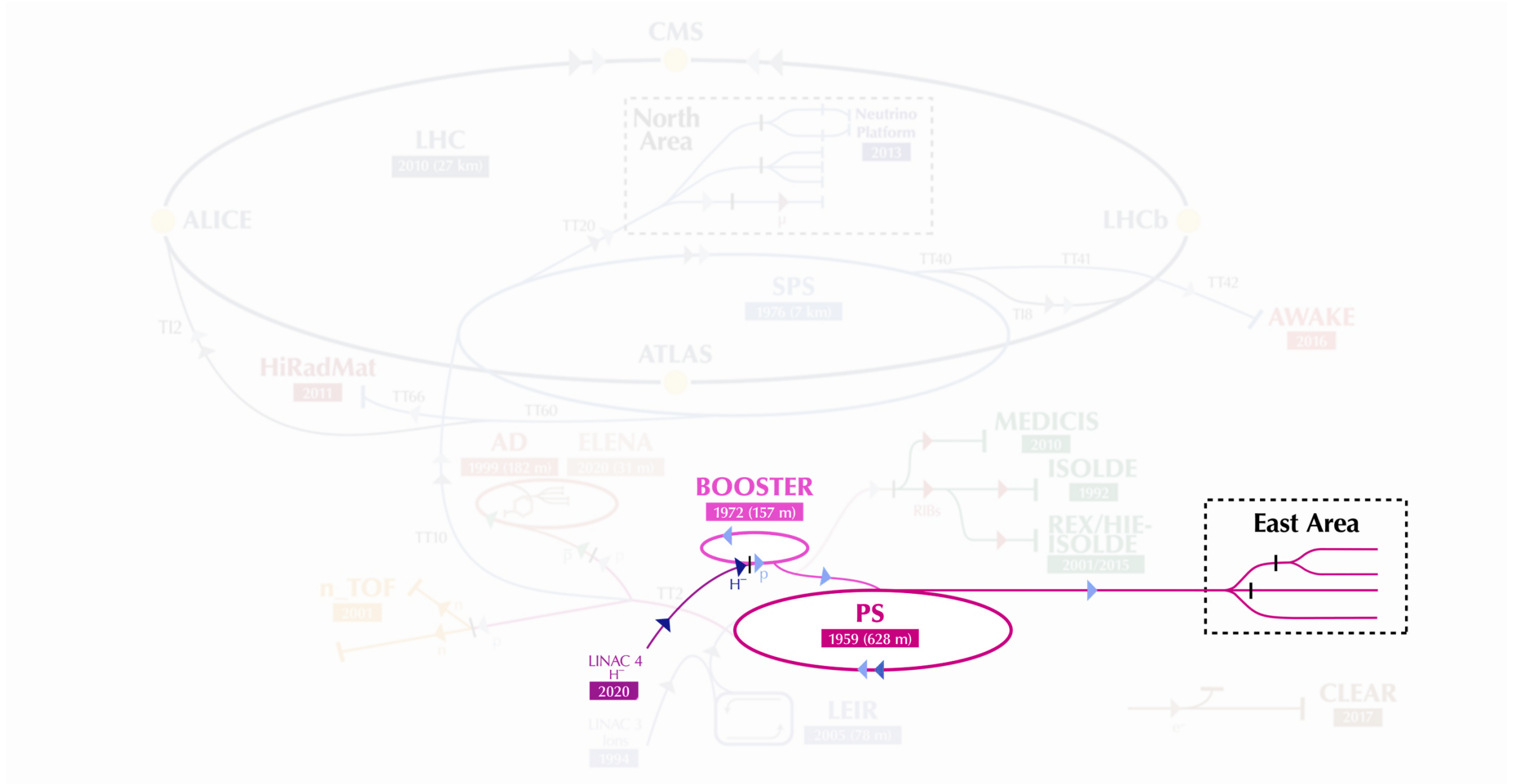
CERN accelerator complex & facilities



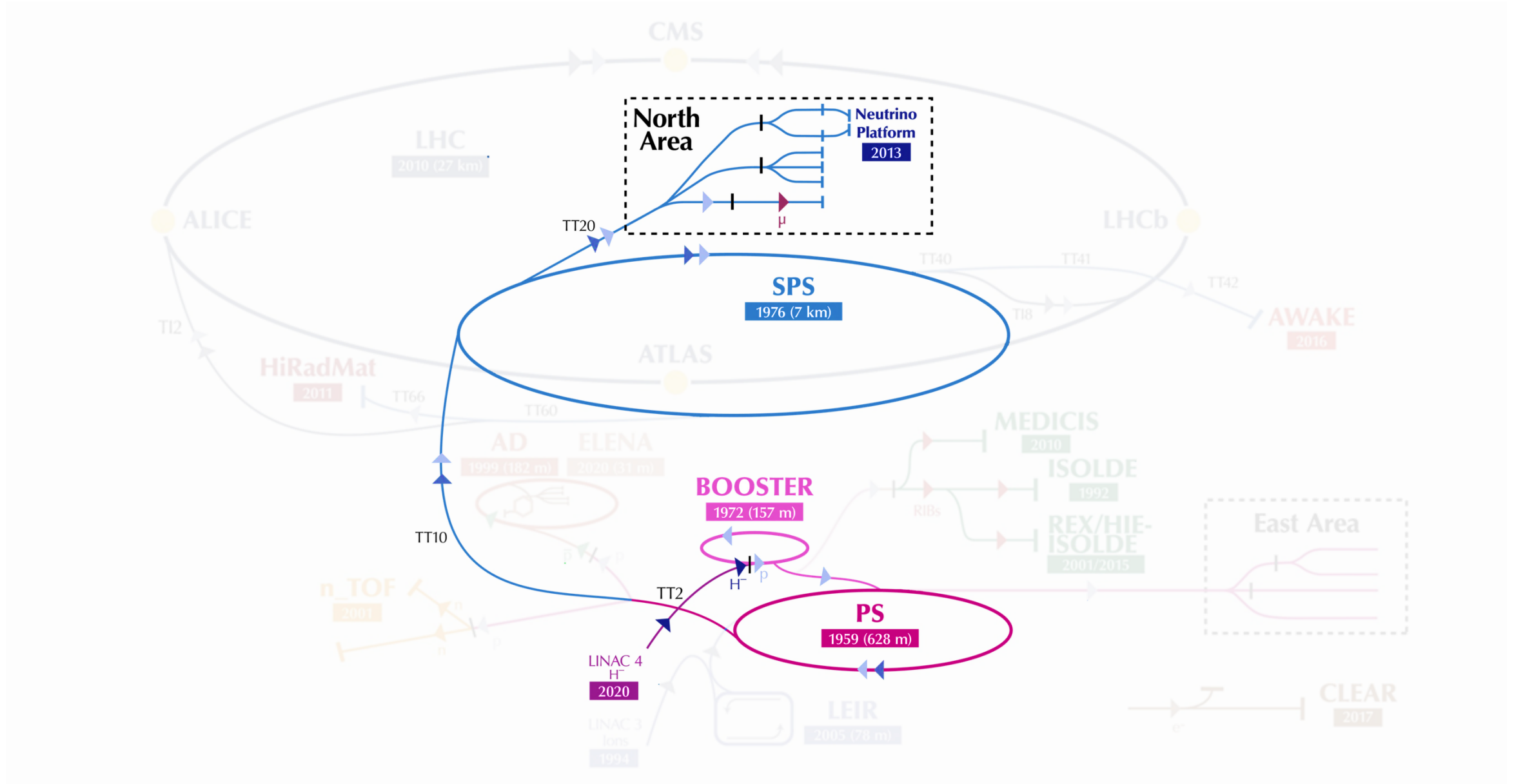
CERN accelerator complex & facilities



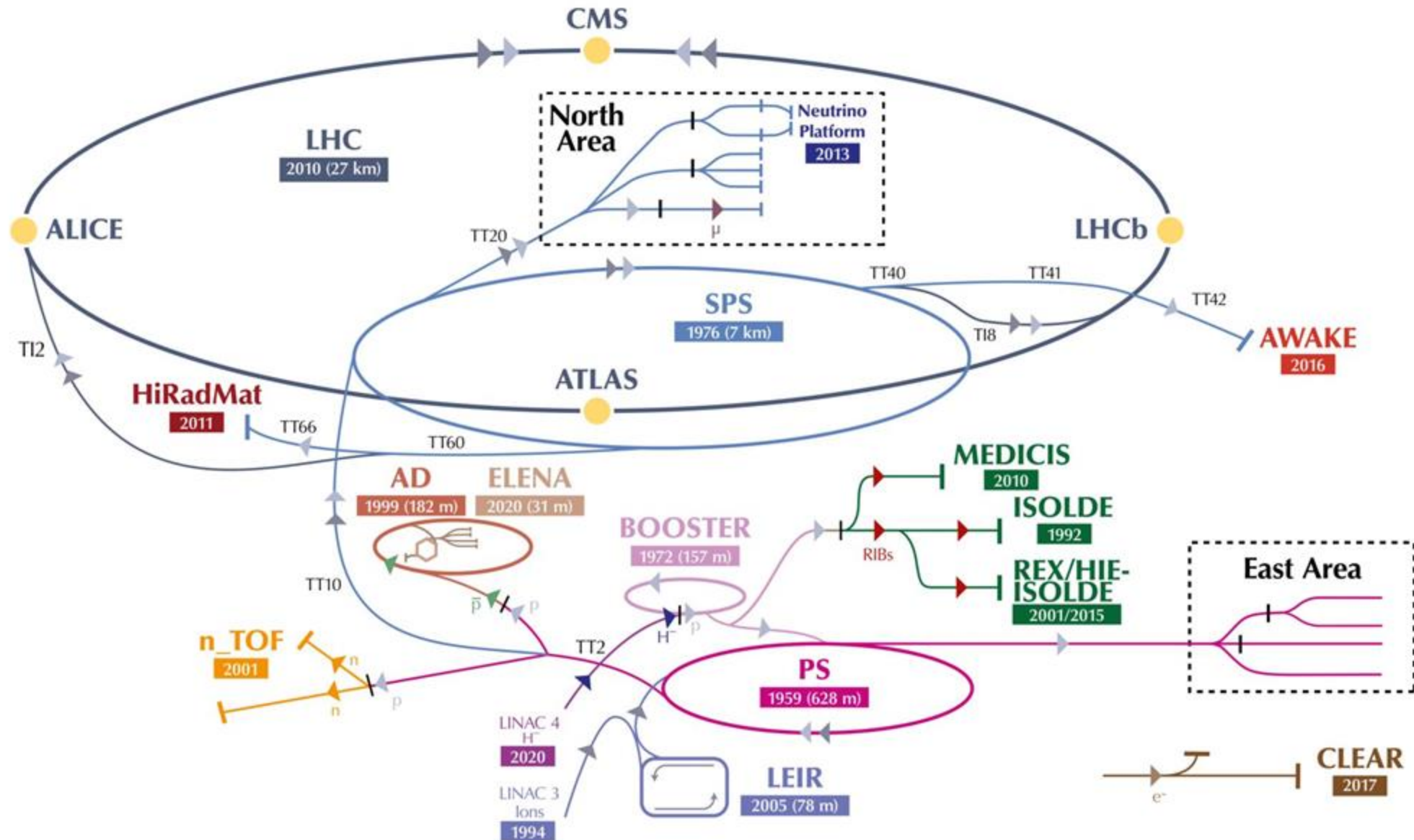
CERN accelerator complex & facilities



CERN accelerator complex & facilities

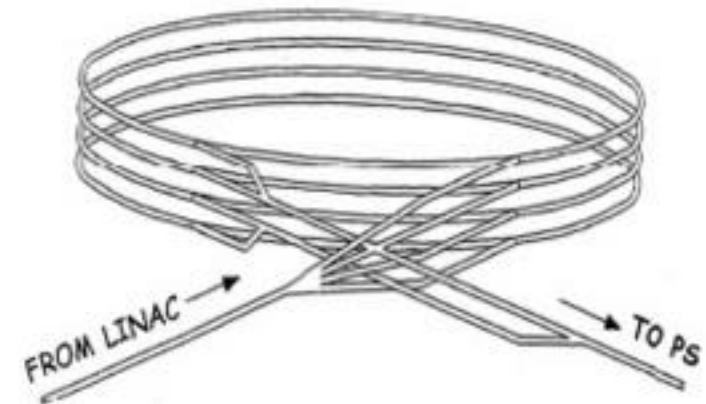
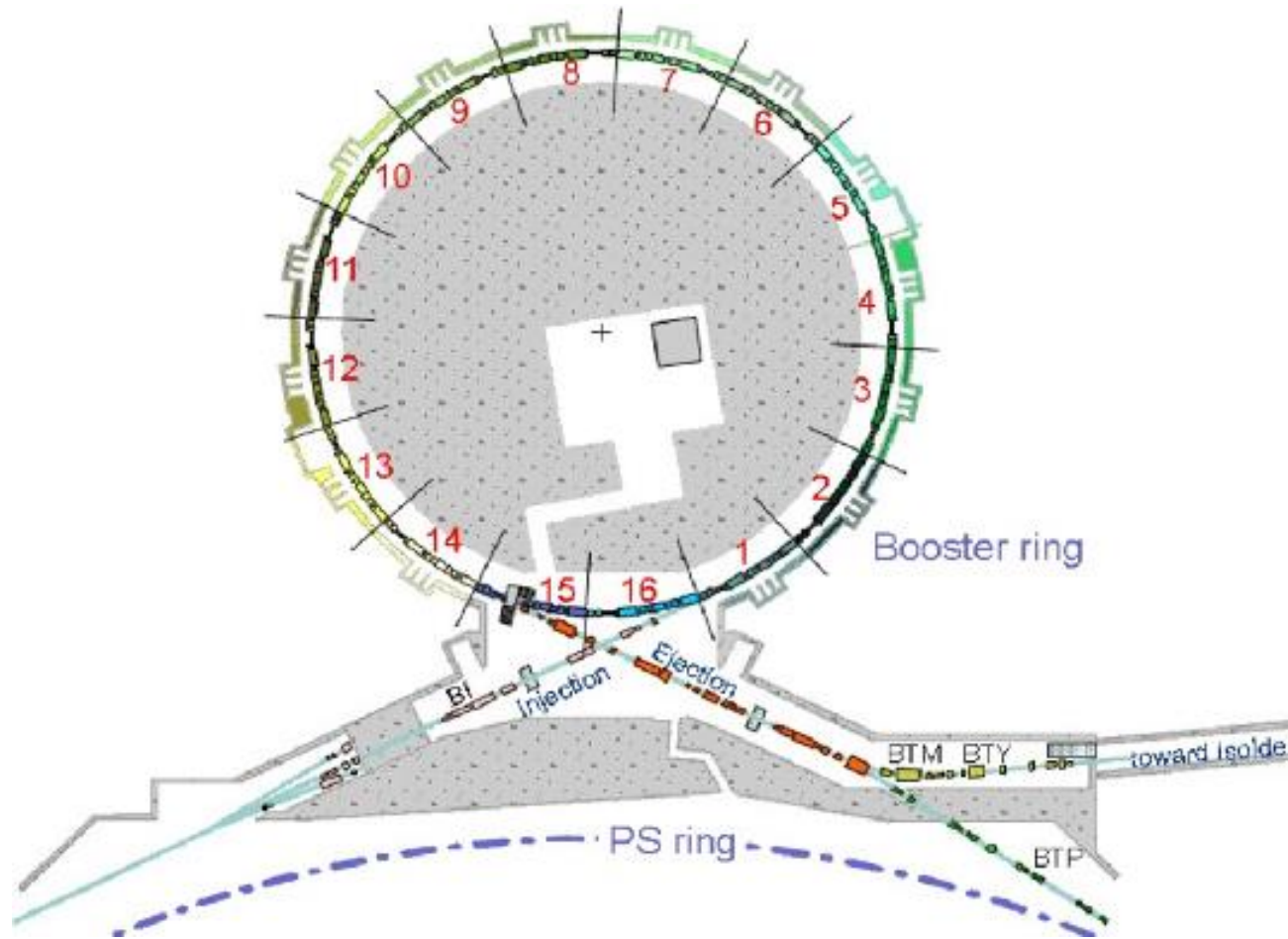


CERN accelerator complex & facilities

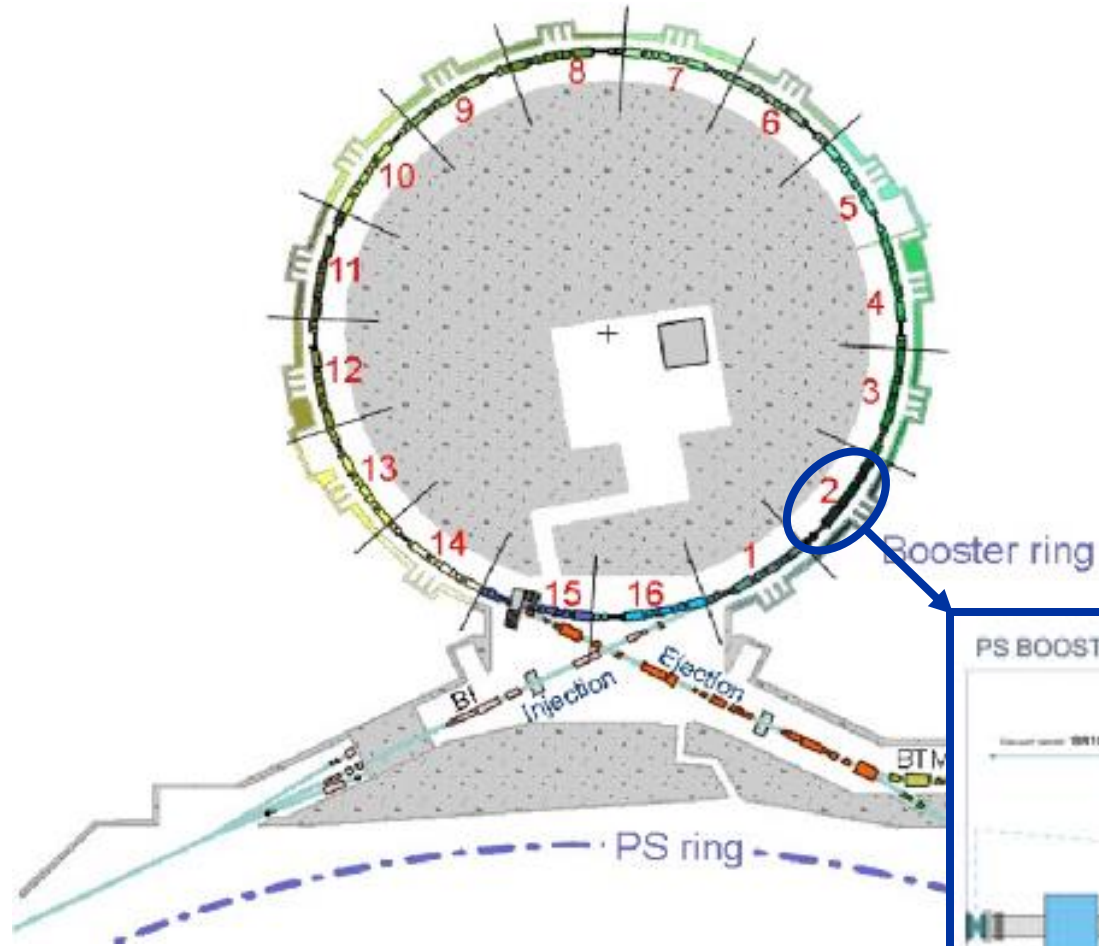


PSB layout

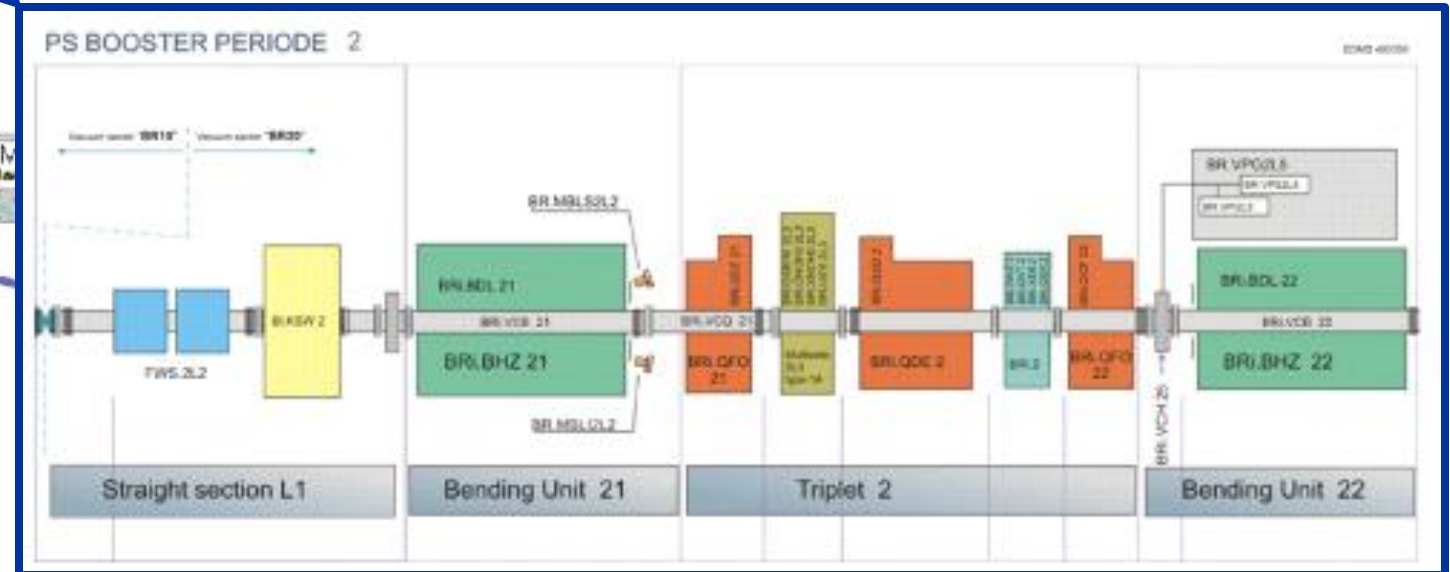
- 4 identical superimposed rings of 25m radius.
- Acceleration from 160 MeV to 2 GeV in ~0.5 sec.
- Common injection and extraction line.



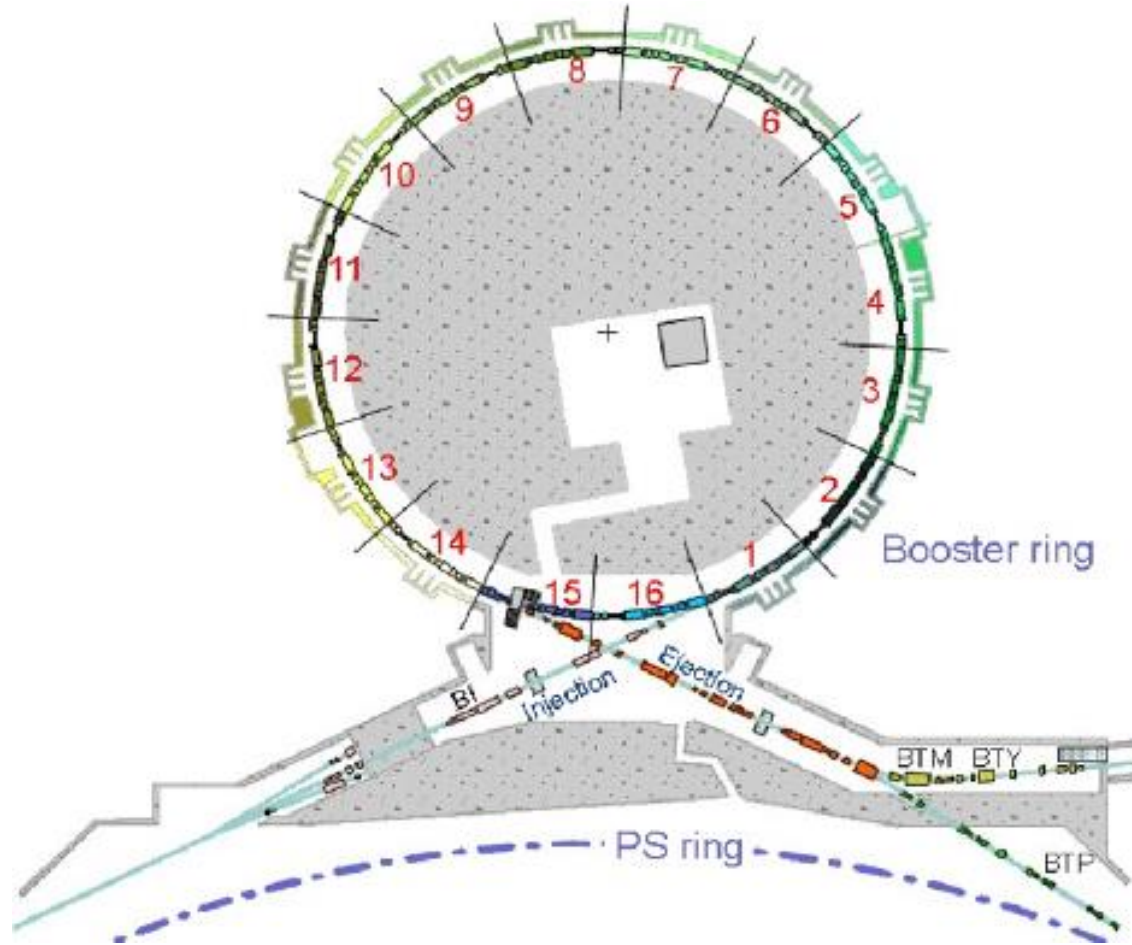
PSB layout



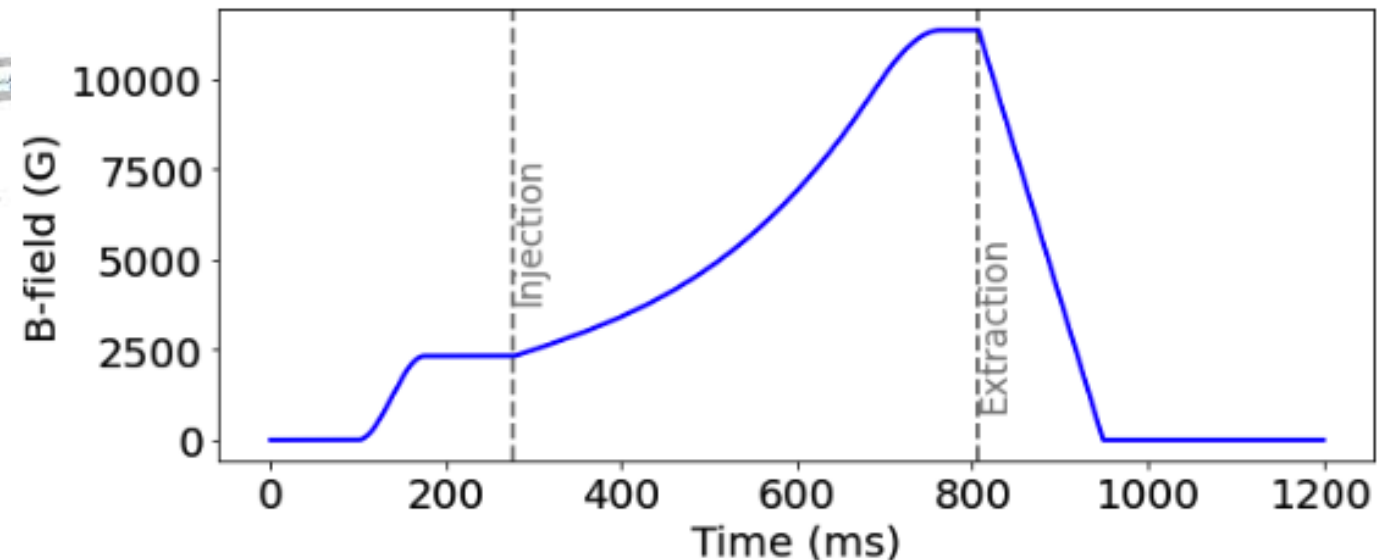
- 4 identical superimposed rings of 25m radius.
- Acceleration from 160 MeV to 2 GeV in ~0.5 sec.
- Common injection and extraction line.
- 16 periods (sectors), each with **two dipoles**, **three quadrupoles** and a **straight section** for additional elements.



PSB layout



- 4 identical superimposed rings of 25m radius.
- Acceleration from 160 MeV to 2 GeV in ~0.5 sec.
- Common injection and extraction line.
- 16 periods (sectors), each with **two dipoles**, **three quadrupoles** and a **straight section** for additional elements.
- Magnetic cycle of 1.2 sec. Beam injection at 275 ms and extraction at 805 ms.



Beam delivery – supercycles

Typical PSB Beams							
Type	N_b [10^{10} ppb]	ϵ_x [μm]	ϵ_y [μm]	ϵ_δ [eVs]	h	...	Final destination
LHC25	165	< 2.2	< 2.2	1.3	1	...	LHC
BCMS25	85	< 1.2	< 1.2	0.9	1	...	LHC
EAST	170	1 – 2	1 – 2	< 1.3	1	...	East Area (PS)
STAGISO	200-300	< 5	< 4	< 1.6	1	...	ISOLDE (PSB)
AD	400	9	5	1.3	1	...	AD (PS)
SFTPRO/MTE	< 600	$< 6 - 8$	< 4	< 1.3	2	...	North Area (SPS)
NORMGPS/HRS	800	< 15	< 8	< 1.8	1	...	ISOLDE (PSB)
TOF	900	12	8	1.7	1	...	nTOF (PS)
PSB MD	depending on needs						Beam Dump (PSB)

Beam delivery – supercycles

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TOF	900	12	8	1.7	1	...	nTOF (PS)
PSB MD	depending on needs						Beam Dump (PSB)

user (cycle name) *mapped* *selector* *added* *supercycle*
(sequence of cycles)

In practice: **LHC25A_8b4e_2025** $\longrightarrow$ **PSB.USER.LHC2A** $\longrightarrow$

MD15243_EUROLABS_2025 $\longrightarrow$ **PSB.USER.MDX** $\longrightarrow$

BP	User
1	MTE_2025
2	---ZERO---
3	ISOGPS_2025
4	EAST_N_2025
5	ISOGPS_2025
6	TOF_2025
7	---ZERO---
8	ISOGPS_2025

BP	User
1	MTE_2025
2	--ZERO--
3	ISOGPS_2025
4	EAST_N_2025
5	ISOGPS_2025
6	TOF_2025
7	--ZERO--
8	ISOGPS_2025
9	TOF_2025
10	ISOGPS_2025
11	LHCINDIV_HiRadM
12	ISOGPS_2025
	--ZERO--

Beam delivery – supercycles

PSB Fixdisplay - W 41

10-Oct-2022 10:30:32

Comments (07-Oct-2022 11:53:24)

Supervisor : S. Albright 167748

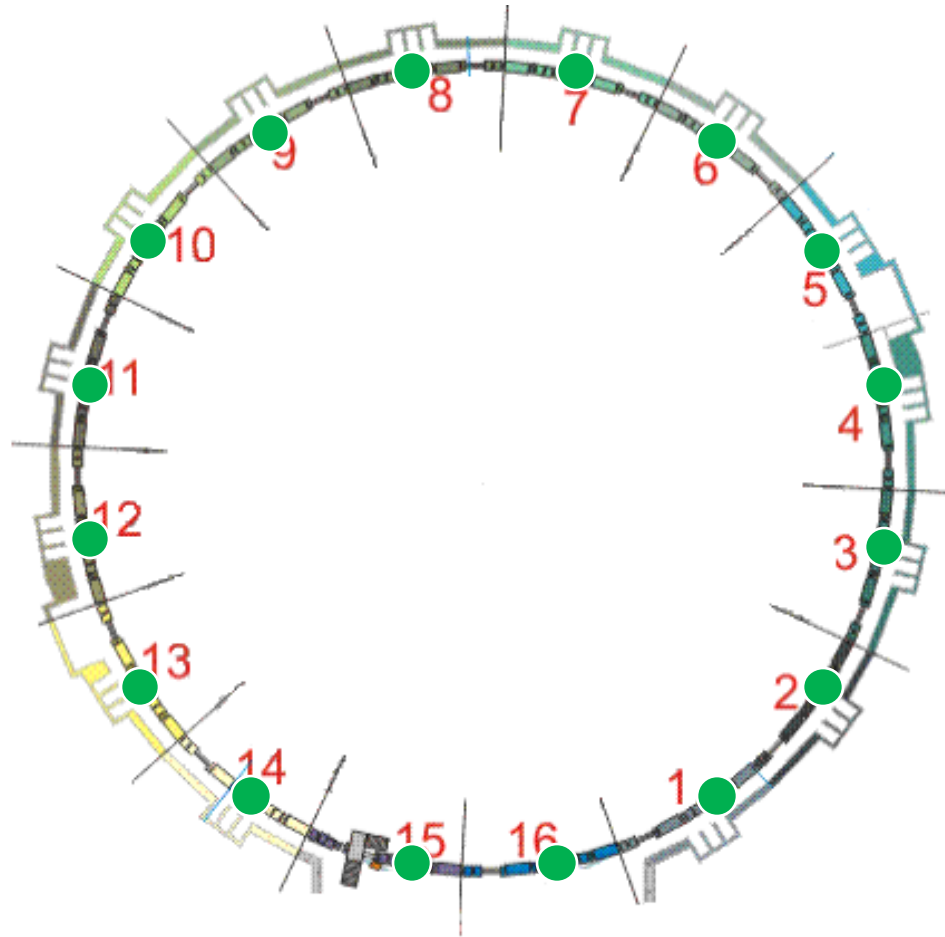
Operator : CCC: 76671

BP	User	Pls	Inj.	Acc.	b.Ej.E10	Ej.E10	Dest.
23	---ZERO---	1	○○○○	○○○○	0.00	0.34	BDUMP
24	---ZERO---	1	○○○○	○○○○	0.00	0.07	BDUMP
25	EAST_T8_2022	2	○○●○	○○●○	59.88	60.37	EAST_T8_22
26	---ZERO---	1	○○○○	○○○○	0.00	0.47	BDUMP
27	---ZERO---	1	○○○○	○○○○	0.00	0.15	BDUMP
1	MTE_2022_EM	21	●●●●	●●●●	2464	2474	MTE_22
2	MTE_2022_EM	21	●●●●	●●●●	2465	2461	MTE_22
3	ISOGPS_2022	18	●●●●	●●●●	1635	1611	BDUMP
4	---ZERO---	1	○○○○	○○○○	0.00	0.37	BDUMP
5	EAST_T8_2022	2	○○●○	○○●○	60.17	61.46	EAST_T8_22
6	---ZERO---	1	○○○○	○○○○	0.00	0.20	BDUMP
8	EAST_T9_2022	3	○○○○	○○○○	0.00	0.44	BDUMP
	---ZERO---						BDUMP

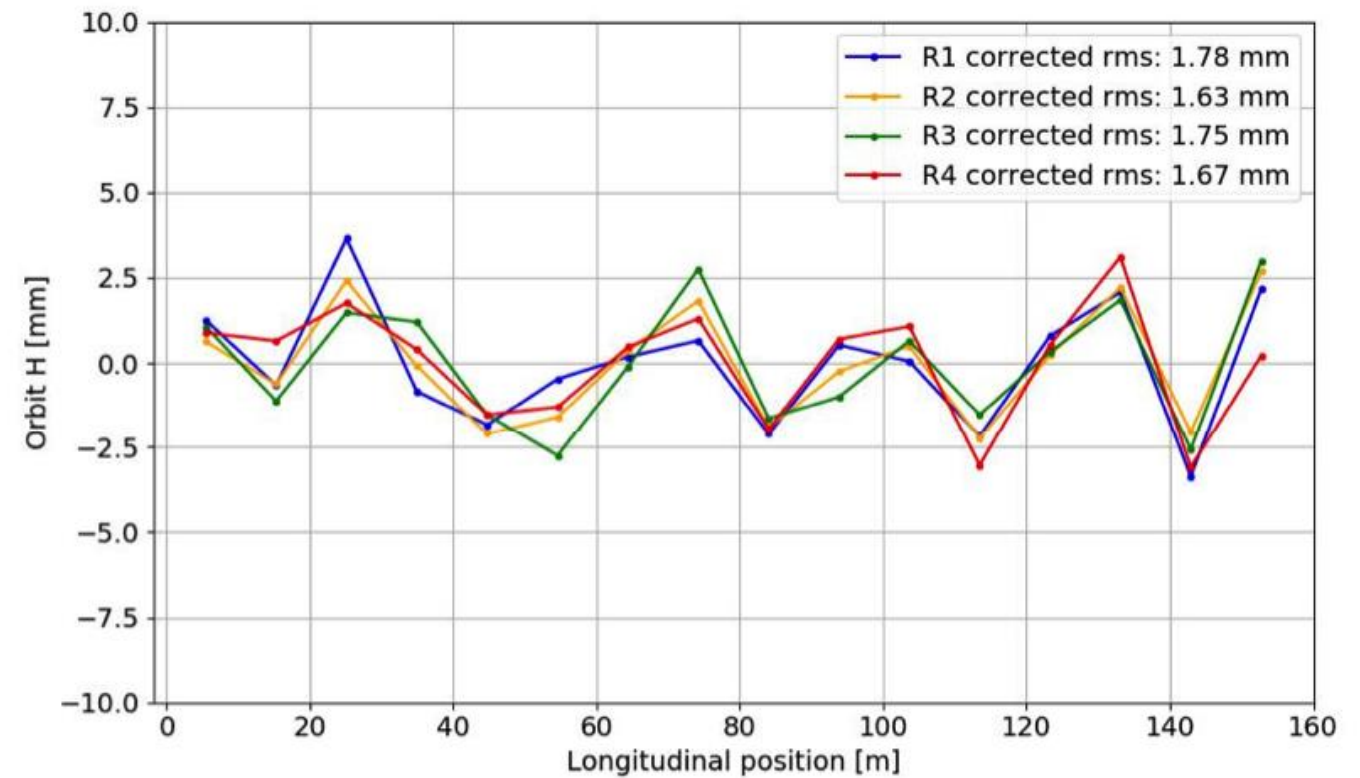
8/27 No Message

- PSB is constantly cycled with different beams, each with its own characteristics and destination.
- The cycles follow a predefined sequence (**supercycle**) which is repeated many times.

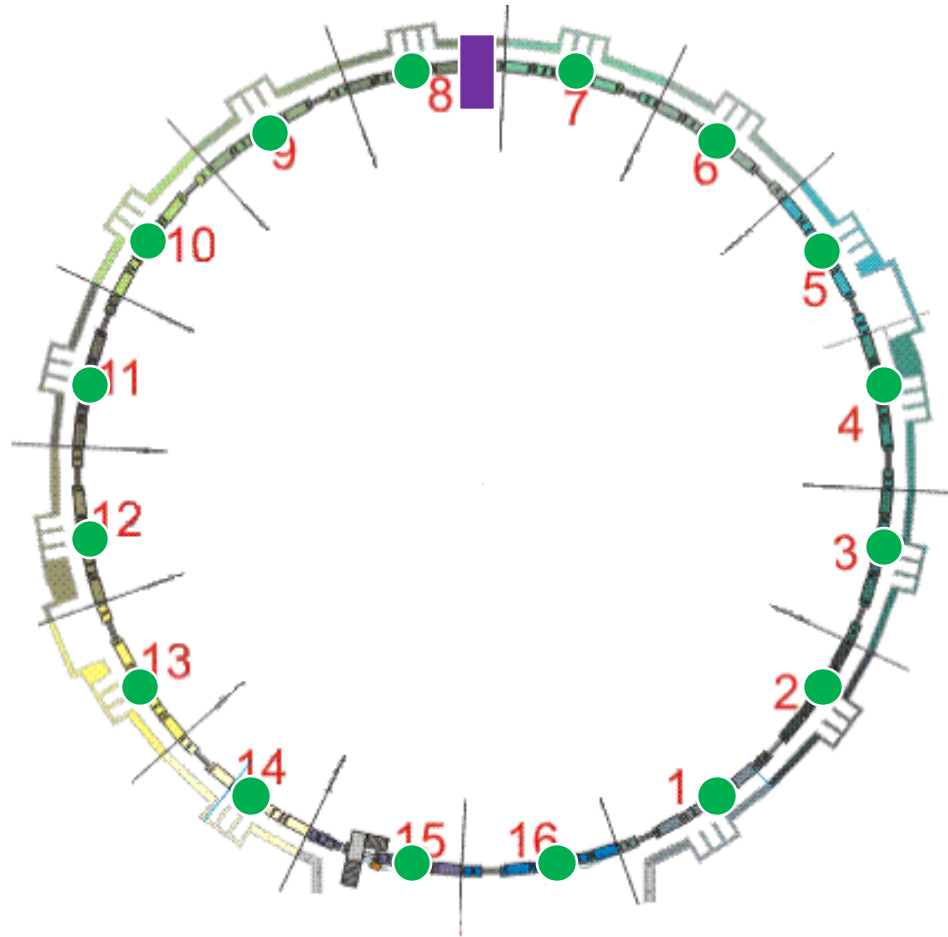
PSB beam instrumentation



- For each of the 4 rings, 16 horizontal and 16 vertical **Beam Position Monitors (BPM)** measure the average transverse position of the beam.



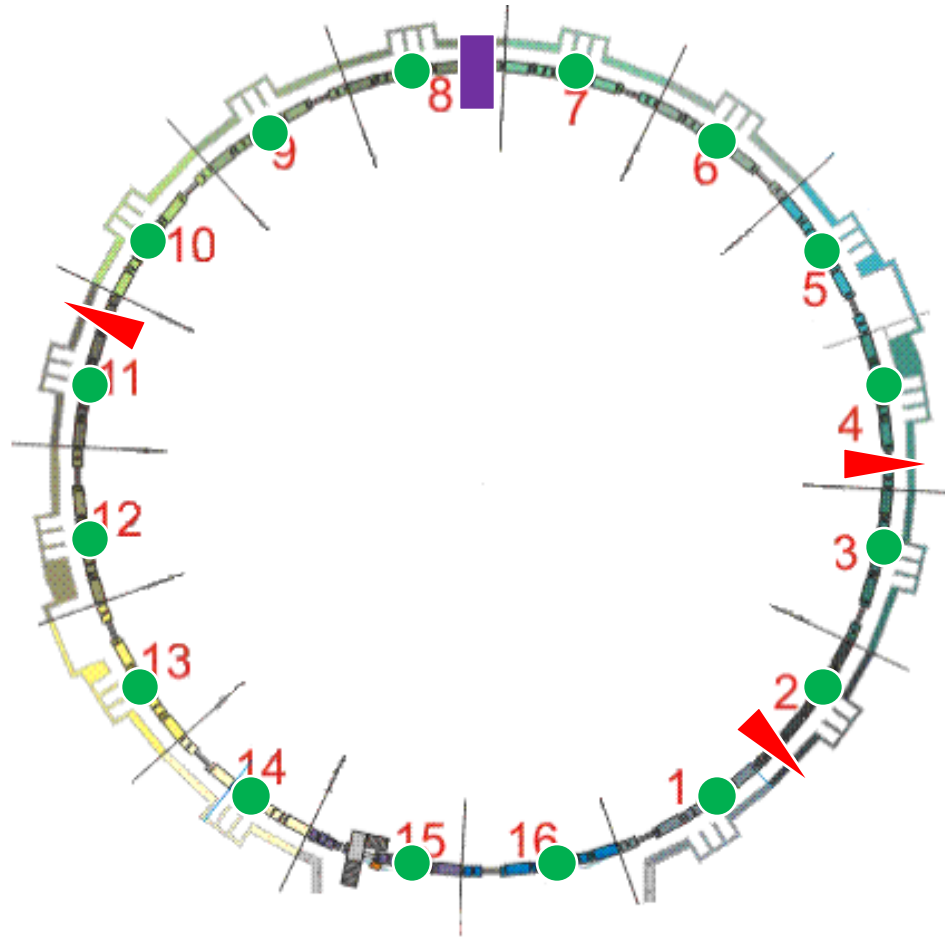
PSB beam instrumentation



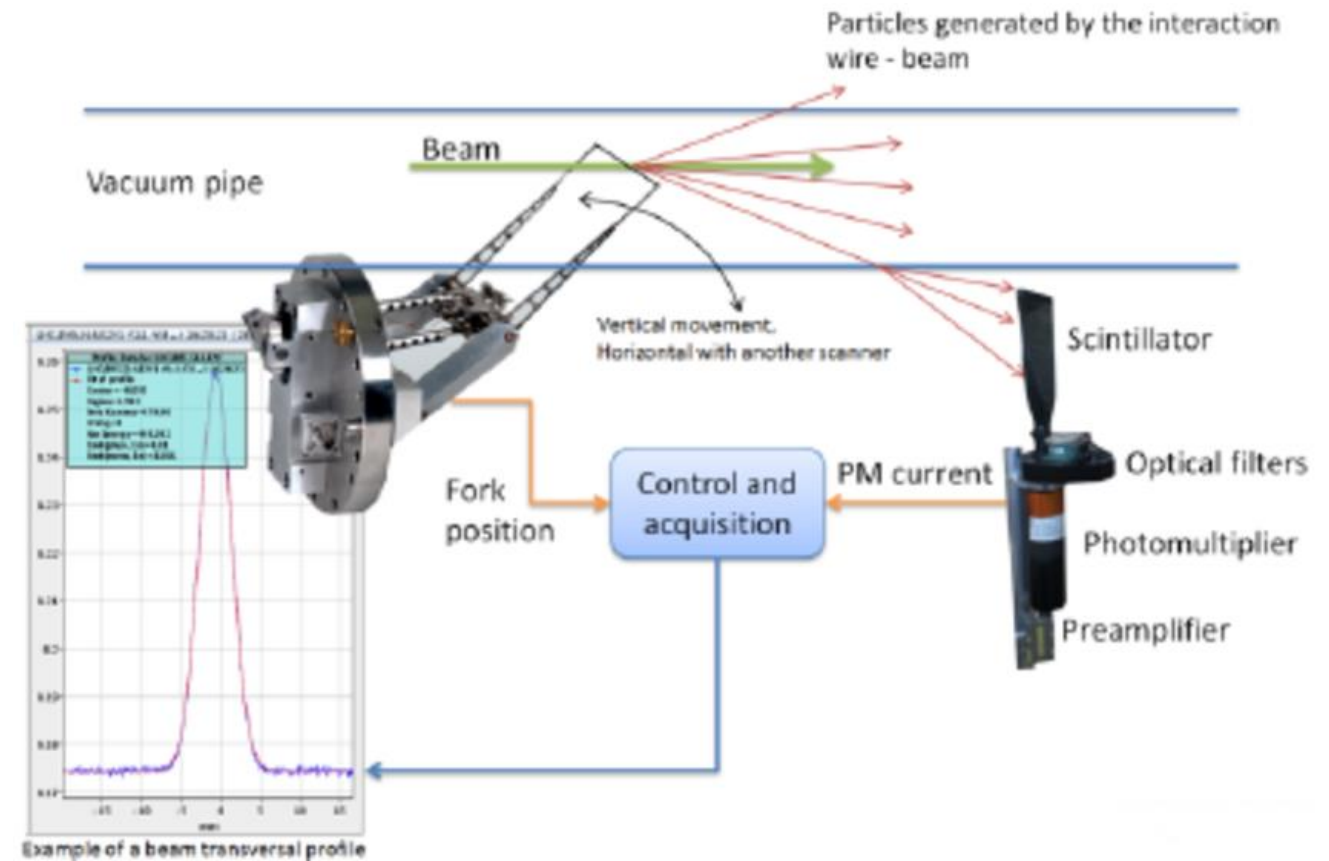
- A **Beam Current Transformer (BCT)** measures the beam intensity (or number of particles) with a 1 ms resolution (~500 points).



PSB beam instrumentation

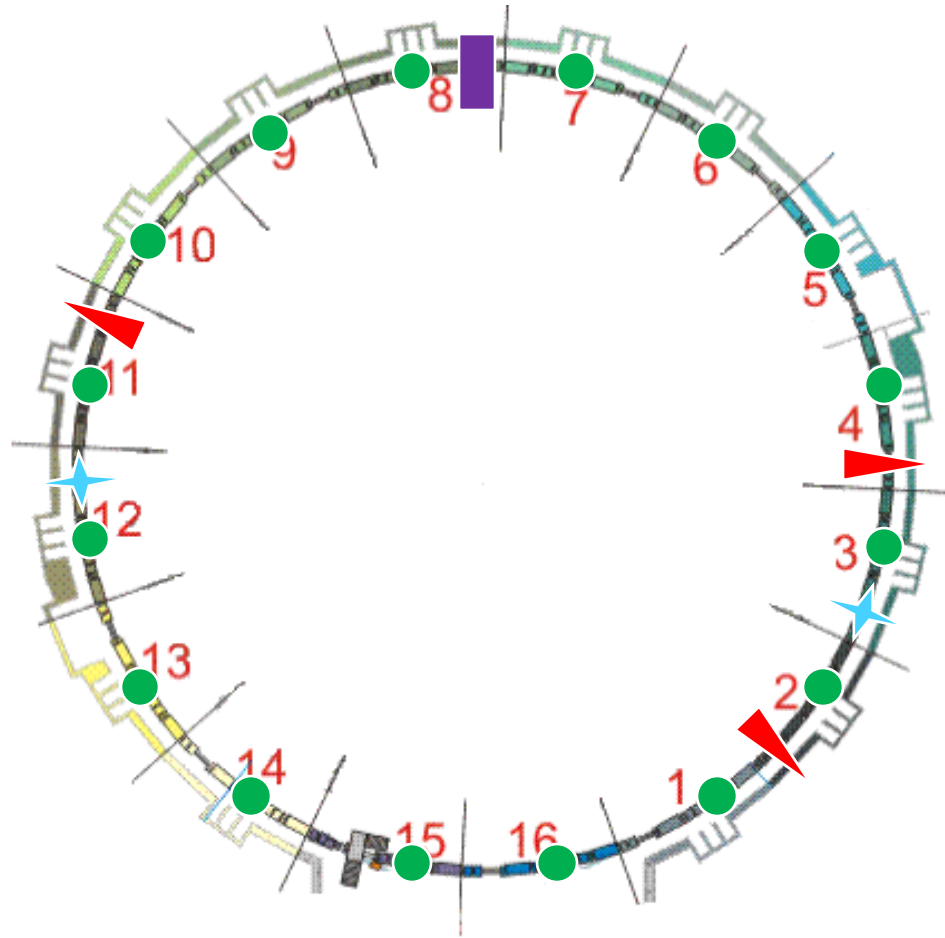


- The beam profile is measured with **Wire Scanners**. The emittance can be calculated from the beam size and the optical functions from the model.

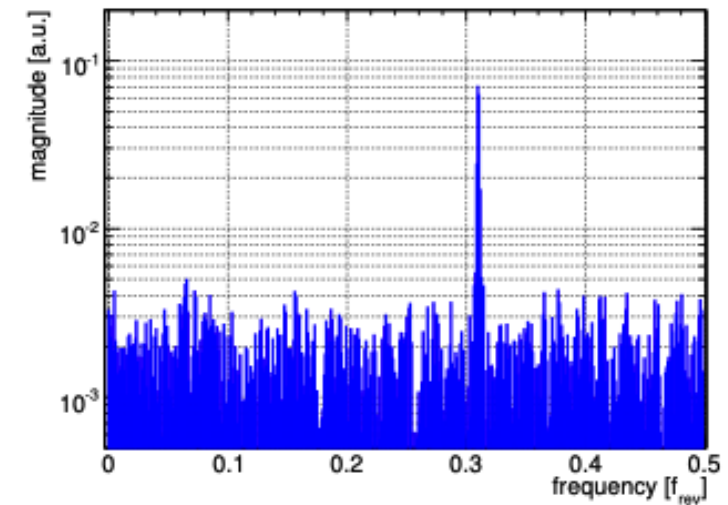
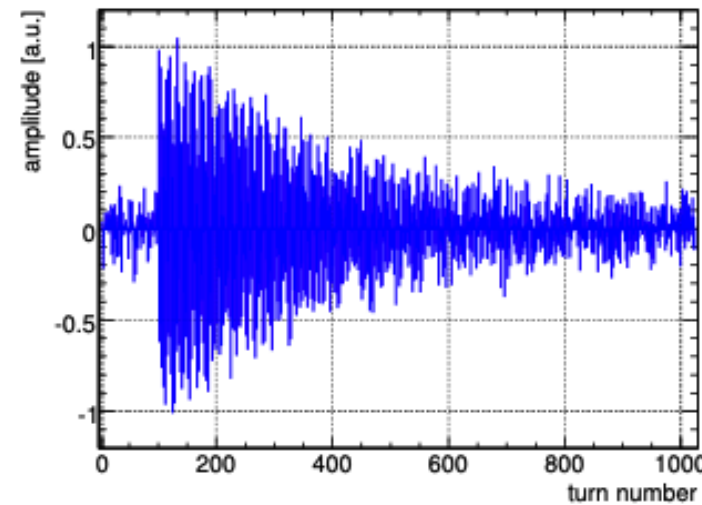


$$\epsilon_{x,y} = \frac{\beta_r \gamma_r}{\beta_{x,y}} \left[\sigma_{x,y}^2 - D_{x,y} \left(\frac{\delta p}{p} \right)_{RMS} \right]$$

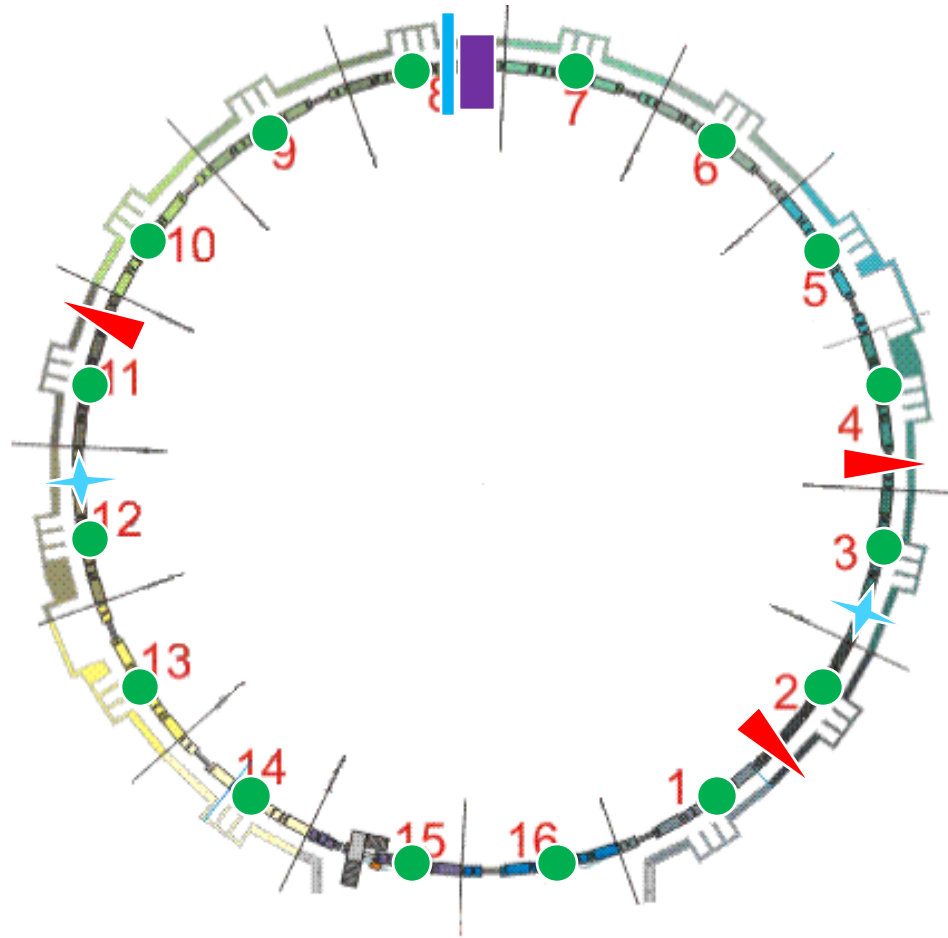
PSB beam instrumentation



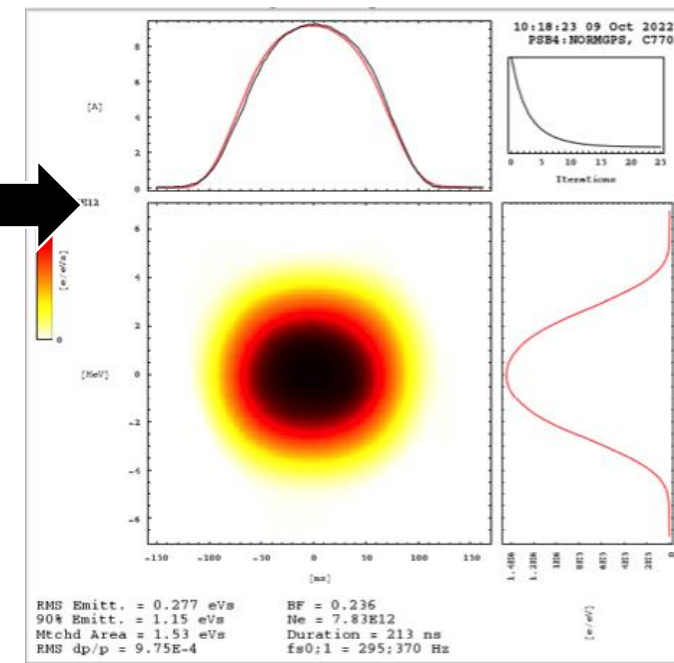
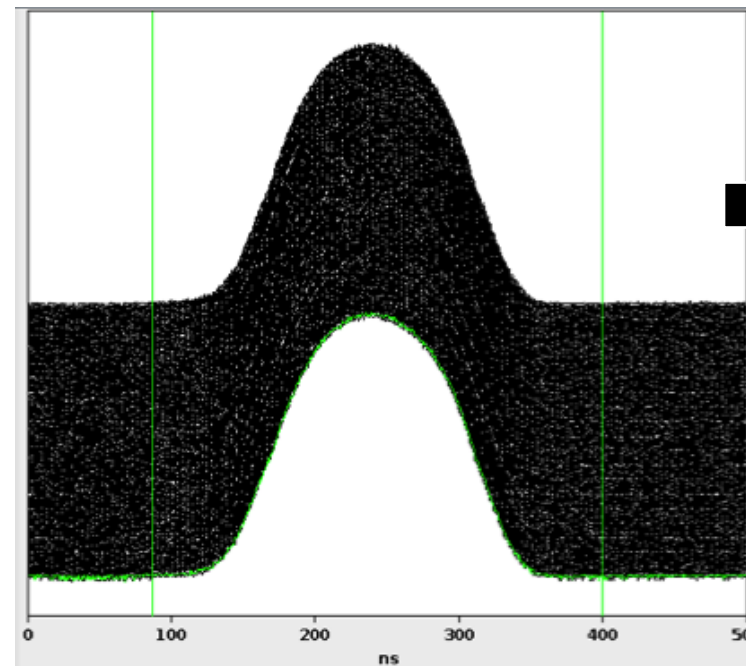
- The tune is measured with a **Base-Band Tune (BBQ)**: a feedback system excites transverse beam oscillations using a frequency sweep (“chirp”), and a pickup records turn-by-turn beam position data. The fractional tune is determined with a Fast-Fourier-Transform (FFT).



PSB beam instrumentation

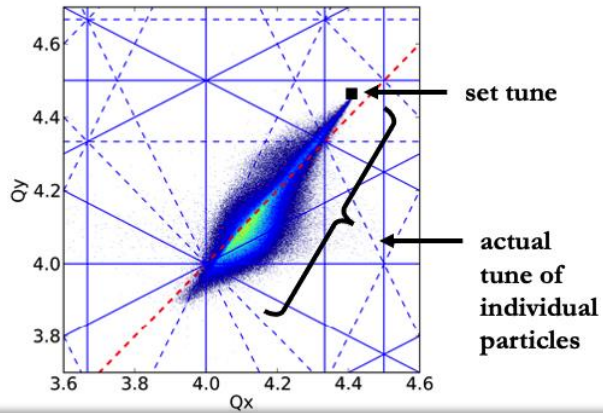


- **Tomoscope**: longitudinal phase space reconstruction. The longitudinal profile measurements over many turns correspond to different projections of the phase space at different angles (due to the synchrotron motion). By properly combining these profiles, the 2-D longitudinal phase space is reconstructed (like tomography).

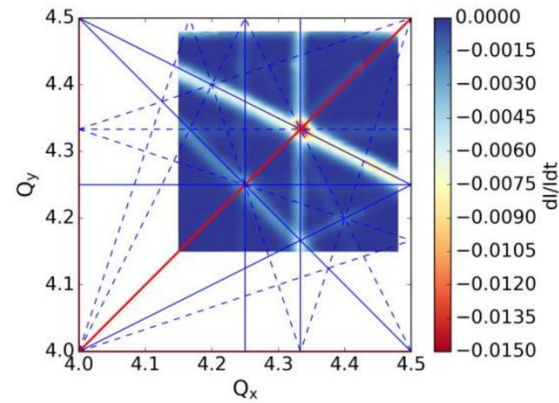


Relevant beam dynamics in the PSB

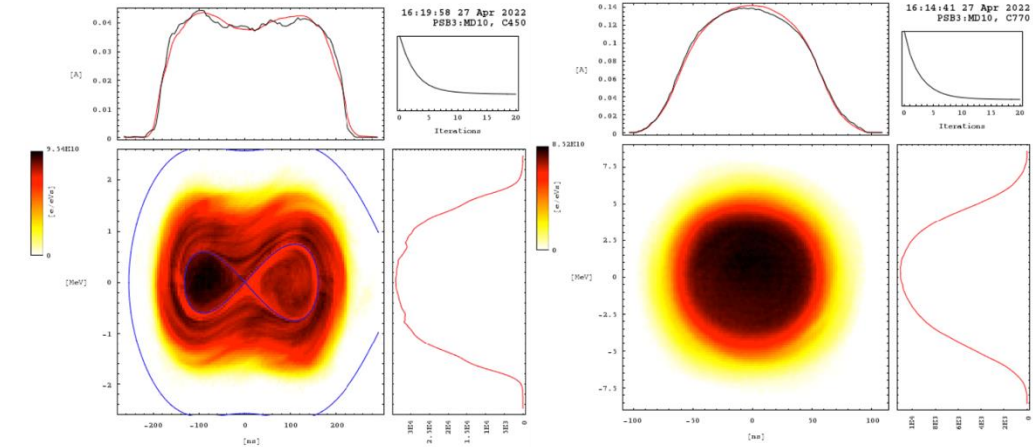
Space Charge



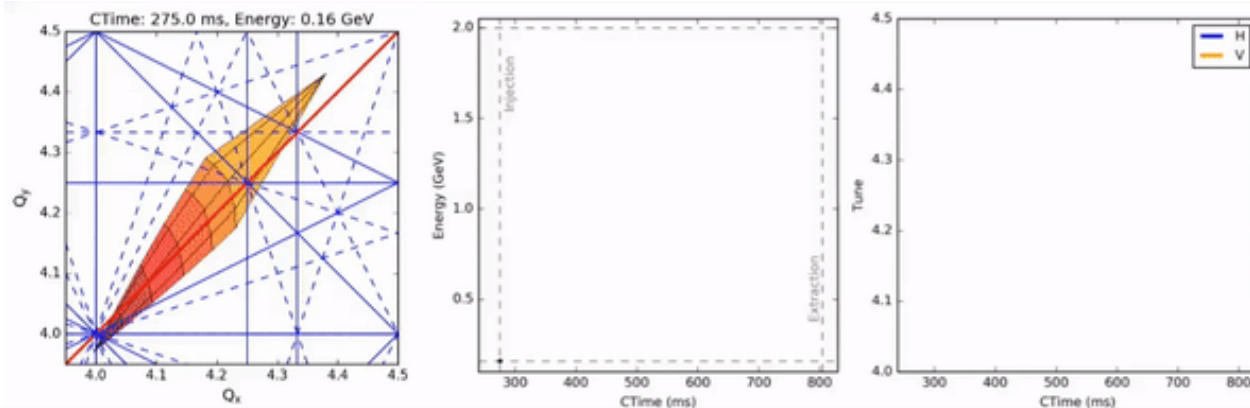
Resonances



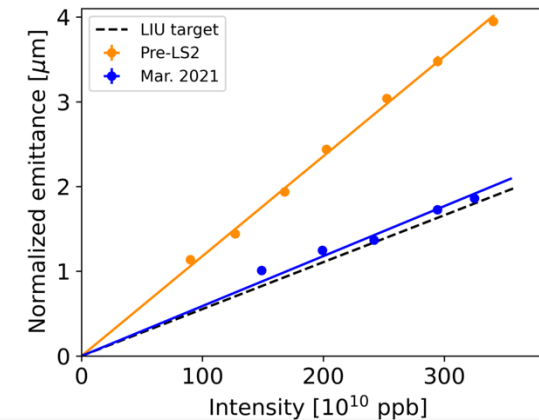
Beam goes from double harmonic to single harmonic to reduce the peak line density (therefore space charge)



While beam is accelerated, tunes are dynamically changed.



Beam Brightness



$$B = \frac{I}{0.5(\epsilon_x + \epsilon_y)}$$

Data acquisition

```
# import PyJapcScout
from pyjapcscout import PyJapcScout
# some other imports for user use
import glob
import numpy as np

myPyJapc = PyJapcScout(incaAcceleratorName='PSB')
myPyJapc.rbacLogin()
mySelector = 'PSB.USER.MD9'
myPyJapc.setDefaultSelector(mySelector)

signalsToMonitor = [
    'BR1.BWS.4L1.H_UCAP/BestChannelResultSet1',
    #'BR2.BWS.4L1.H_UCAP/BestChannelResultSet1',
    #'BR3.BWS.4L1.H_UCAP/BestChannelResultSet1',
    #'BR4.BWS.4L1.H_UCAP/BestChannelResultSet1',

    'BR1.BWS.11L1.V_UCAP/BestChannelResultSet1',
    #'BR2.BWS.11L1.V_UCAP/BestChannelResultSet1',
    #'BR3.BWS.11L1.V_UCAP/BestChannelResultSet1',
    #'BR4.BWS.11L1.V_UCAP/BestChannelResultSet1'
]

shots=30
def myCallback(data, h):
    global shots
    print( 'Shot ' + str(len(glob.glob(h.saveDataPath + '/2025*'))))
    indx = len(glob.glob(h.saveDataPath + '/2025*'))
    if indx == shots:
        h.stopMonitor()
        print('Measurement finished and monitor stopped.')

myMonitor = myPyJapc.createMonitor(signalsToMonitor, onValueReceived=myCallback,
```

- Accelerator and beam data can be retrieved either from the **accelerator logging system** ([NXCALS](#)) or recorded locally using **libraries for interacting with the accelerator control system** (e.g. [PyJAPC](#), [PyJAPCScout](#), ...)

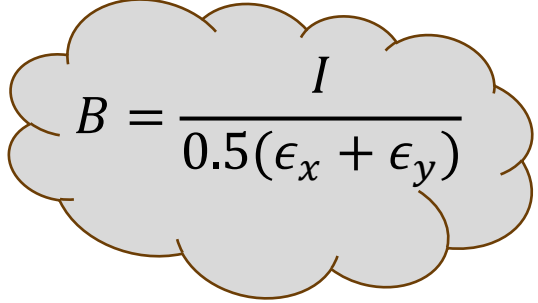
Measurements of today

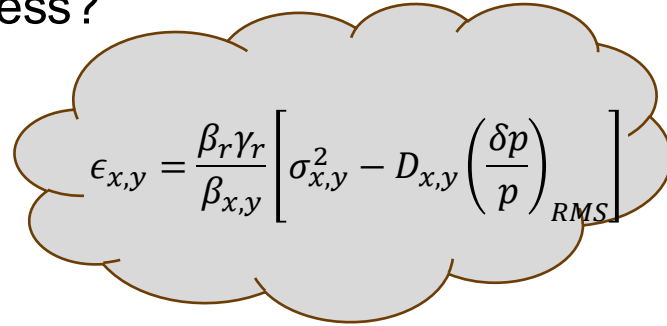
Getting familiar

- Beam characterization
 - Get familiar with the PSB and basic instruments (*BCT*, *Wire Scanner*, *Tomoscope*).
 - Monitor basic beam parameters (beam intensity, emittance, bunch shape)
- Injection effects
 - Understand multi-turn injection system and how to control it (*Cruise Control*).
 - Optimize injection process (minimize injection oscillations and foil scattering)
- Tune setting and measurement
 - Get familiar with LSA interface.
 - Perform tune measurement (and correction if needed) all along the cycle for various working points.

PSB brightness characterization

- Start with a single bunch in a 160 MeV flat-bottom cycle at the working point of $(Q_x, Q_y) = (4.17, 4.23)$ (one PSB ring). Measure the beam brightness up to the intensity of $I = 150 \times 10^{10}$ protons (e.g. 10, 50, 100, 150).
- Change horizontal tune (e.g. to 4.21, 4.25, 4.29) and repeat.
 - What do you observe in the beam brightness?
- Do the same for the vertical tune (e.g. 4.27, 4.31 and 4.35).
 - Why beam losses occur at the high vertical tunes?
 - How would you optimize the working point to maximize the beam brightness?
- Repeat for 3 different KSW settings at the initial working point.
 - What is the effect of the foil in the beam brightness?
- Repeat the brightness measurement for the initial working point and an unbunched (coasting) beam.
 - What do you observe in the beam brightness?
 - Why can't we use unbunched beams in operation?


$$B = \frac{I}{0.5(\epsilon_x + \epsilon_y)}$$

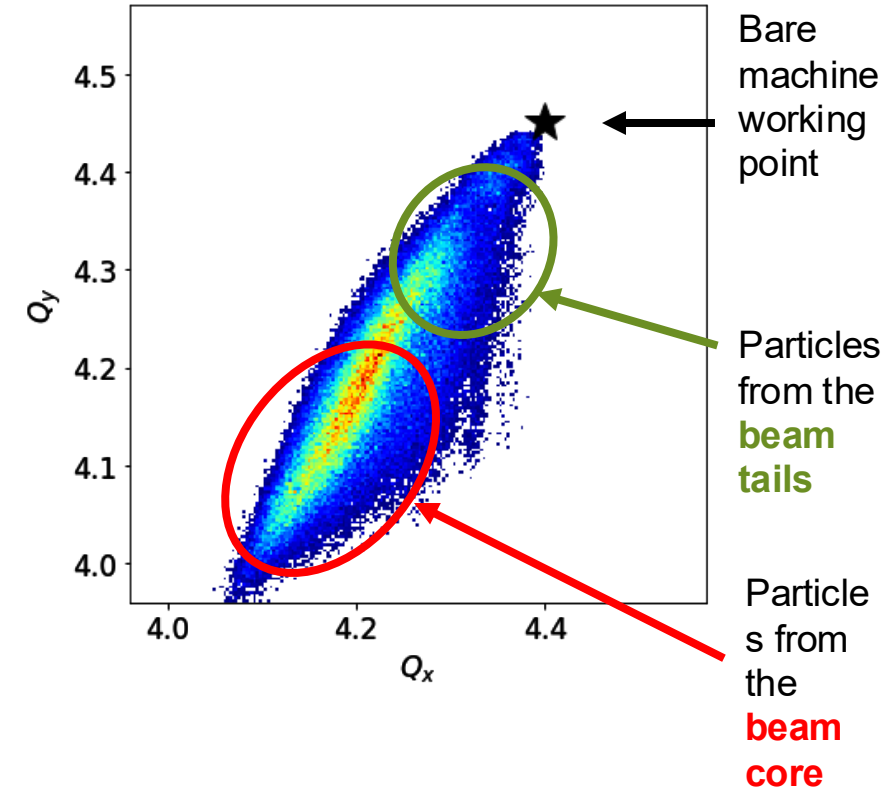
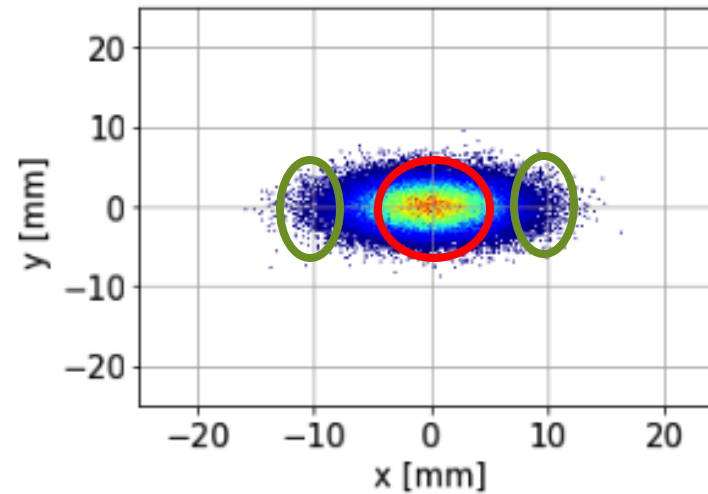

$$\epsilon_{x,y} = \frac{\beta_r \gamma_r}{\beta_{x,y}} \left[\sigma_{x,y}^2 - D_{x,y} \left(\frac{\delta p}{p} \right)_{RMS} \right]$$

Space charge & resonances

Beam dynamics in PSB

PSB performance is dominated by space charge and resonances.

- Direct space charge induces a spread in the tune of the particles.
 - Defocusing depends on the particle's amplitude: **larger tune shift** for the particles in the **beam core**.



Beam dynamics in PSB

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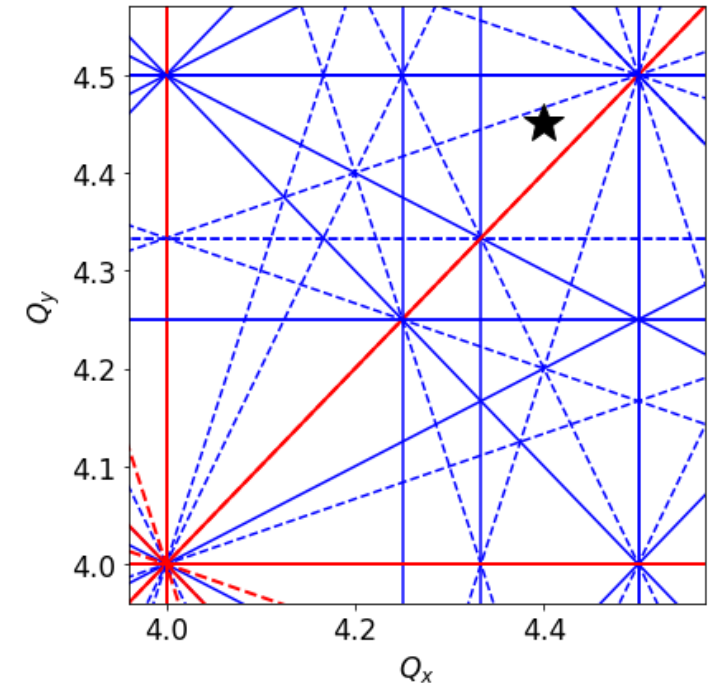
- Direct space charge induces a spread in the tune of the particles.
 - Defocusing depends on the particle's amplitude: **larger tune shift** for the particles in the **beam core**.
- Machine imperfections (field perturbations, misalignments, ...) can excite resonances.
 - Resonances are identified and corrected experimentally using available multipole correctors.

Resonance condition:

$$mQ_x + nQ_y = p, \quad m, n, p \in \mathbb{Z}$$

Resonances are represented with lines in the tune diagram.

Motion near a resonance can become unstable and particles might get lost! Need to **tune the machine in a “resonance free”** region.

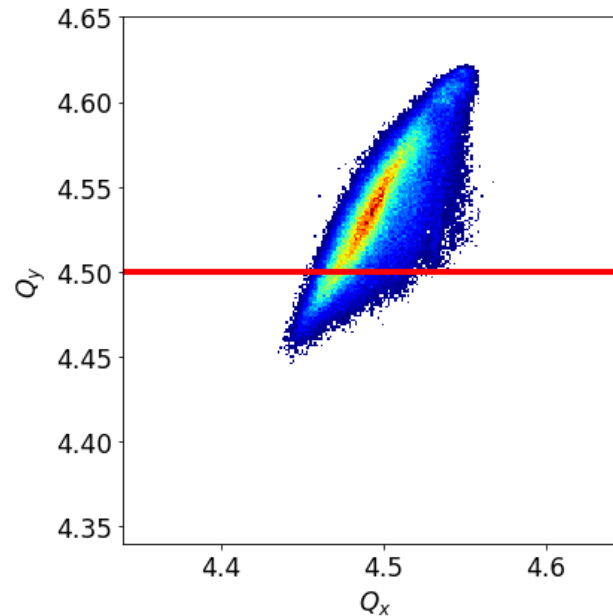


Beam dynamics in PSB

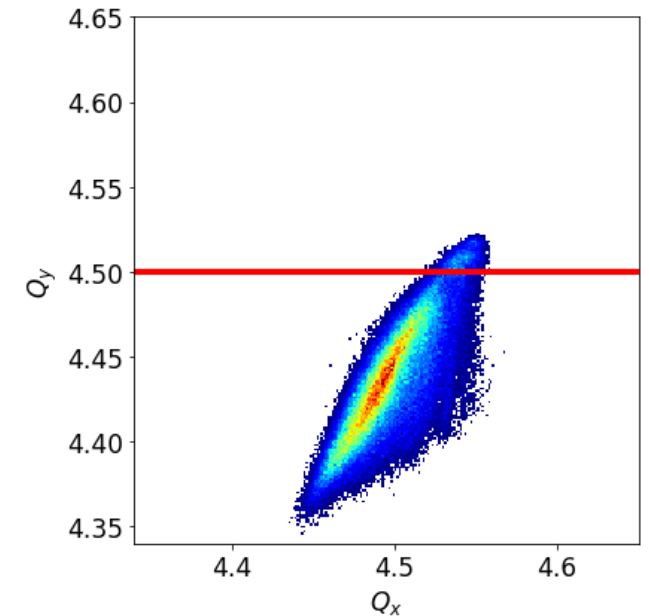
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- Space charge detuning can drive particles to resonances.

Low amplitude particles (beam core) interact with the resonance: **emittance blow-up**



High amplitude particles (beam tails) interact with the resonance: **losses**



Beam dynamics in PSB

PSB performance is dominated by space charge and resonances.

- Direct space charge induces a spread in the tune of the particles.
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- Machine imperfections (field perturbations, misalignments, ...) can excite resonances.
 - Resonances are identified and corrected experimentally using available multipole correctors.
- Space charge detuning can drive particles to resonances.
- **The PSB beam brightness is limited by space charge.**

For high beam intensities (high space charge), the detuning is so large that the emittance blows-up by the **integer resonances**.

The higher the intensity, the higher the emittance: **constant brightness**.

